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**MI-302 (GS)**  
**B.E. III Semester Examination, June 2020**  
**Grading System (GS)**  
**Mechanics of Solid and Fluids**  
**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Define the terms : Poisson's Ratio, Strain measurement, Principle planes.  
b) Explain the meaning of the terms tensile stress and thermal stress in a bar.
2. A 45kg block of aluminium is hung from the end of a vertical 0.50m long steel wire with a cross-sectional area of  $5.0 \times 10^{-5} \text{m}^2$ .
  - i) Find the stress and strain in the wire.
  - ii) What is the increase in the length of the steel wire?  
Young's modulus of aluminium :  $7.0 \times 10^{10} \text{ Pa}$   
Young's modulus of steel :  $2.0 \times 10^{11} \text{ Pa}$   
Tensile strength of steel :  $3.0 \times 10^8 \text{ Pa}$
3. What is strain hardening? Discuss briefly with a neat sketch.
4. A rod 160 cm long and of diameter 2.0 cm is subjected to an axial pull of 22 kN. If the modulus of elasticity of the material of the rod is  $2 \times 10^5 \text{ N/mm}^2$ ; determine:
  - i) The stress
  - ii) The strain
  - iii) The elongation of the rod
5. What is Mohr's circle? State its significance.
6. Explain: for fluid
  - i) Density
  - ii) Specific volume
  - iii) Specific weight
  - iv) Specific gravity

OR

Write short notes on :

- a) Thermal stresses
- b) Meta centric height
- c) Types of manometers

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7. A circular rod of 25mm diameter and 500mm long is subjected to a tensile force of 60kN. Determine modulus of rigidity, bulk modulus and change in volume if Poisson's ratio = 0.3 and Young's modulus  $E = 2 \times 10^5 \text{N/mm}^2$ .

OR

What is vortex flow? Explain the various types of vortex flow.

8. Write short note on following (Any two):

- i) Classification of boundary layer
- ii) Momentum correction factor
- iii) Streak Lines
- iv) Path Lines
- v) Kinetic energy correction factor

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